Due: August 19, 2026

1. For the following equations find dy/dx:

   a) $y = (x^2 + 2x + 3)^2$ (check by multiplying out and then differentiating)

   b) $y = (\frac{x^3}{3} + 1)^5 + (\frac{x^2}{2} + 1)^4 + 3$

   c) $y = [(x^2 + 1)^{10} + 1]^8$

   d) $y = \dfrac{[(x^3 + 7)^4 + x]^5}{x^2 + 1}$

2. Find the derivatives y' using the extended power rule:

   a) $y = x^{3/4} + x^{4/3}$

   b) $y = 1/\sqrt{x}$

   c) $y = 1/\sqrt[3]{2x - 3}$

3. Find $\partial y/\partial x_1$ and $\partial y/\partial x_2$ for each of the following functions:

   a) $y = 7x_1 + 5x_1x_2^2 - 9x_2^3$

   b) $y = \dfrac{4x_1 + 3}{x_2 - 2}$

4. Find $\partial f/\partial x$ and $\partial f/\partial y$ for each of the following:

   a) $f(x, y) = x^2 - 5xy - y^3$

   b) $f(x, y) = (x^2 - 3y)(x - 2)$

5. Find the second and third derivatives of the following functions:

   a) $6x^4 - 3x - 4$

   b) $\dfrac{1 + x}{1 - x}$ $(x \neq 1)$

   c) $-16x^3 + 12x^2 + 4x$

6. Find the stationary values (critical values) of the following (check whether relative maxima, or minima, or inflection points) assuming the domain to be the set of all real numbers:

   a) $y = x^2 + 3$

   b) $y = 3x^2 - 6x + 2$

7. Find the stationary values of the following (check whether relative maxima or minima or inflection points), assuming the domain to be the interval $[0, \infty)$:

   a) $y = x^3 - 3x + 5$
   b) $y = \frac{1}{3}x^3 - x^2 + x + 10$
   c) $y = -x^3 + 4.5x^2 - 6x + 6$

8. Find the maximum and the minimum values and points:

   a) $f(x) = x^2 - 5x + 3$, $-\infty < x < +\infty$
   b) $f(x) = x^3 - x$, $-\infty < x < +\infty$

9. Solve the following exercises by the method of Lagrange multipliers:

   a) Minimize the function $x^2 + 3y + 10$; subject to the constraint: $8 - x - y = 0$
   b) Maximize $x^2 + xy - 3y^2$; subject to the constraint: $2 - x - 2y = 0$
   c) Maximize $-2x^2 - 2xy - \frac{3}{2}y^2 + x + 2y$; subject to the constraint: $x + y - \frac{5}{2} = 0$